DNA Fingerprinting in Crime Investigation: Applications, Challenges, and Future Directions in Forensic Biomedical Computing

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***Abstract*—DNA fingerprinting is one of the most transformative techniques in modern forensic science and biomedical computing. By analysing unique genetic patterns present in biological samples such as blood, hair, saliva, semen, and skin cells, forensic experts can reliably identify suspects, victims, and missing persons. Since its discovery by Sir Alec Jeffreys in 1984, the technology has evolved from radioactive multi-locus probe methods to highly automated Short Tandem Repeat (STR) analysis powered by Polymerase Chain Reaction (PCR). This paper surveys the concept, methodology, current forensic DNA technologies, lineage markers, national database frameworks, and the integration of computational tools including artificial intelligence and next-generation sequencing (NGS) in forensic DNA analysis. Key challenges such as sample contamination, privacy concerns, legal admissibility, and database governance are examined. Future directions including rapid on-site DNA profiling, machine learning-assisted interpretation, and federated forensic databases are discussed. The paper concludes that DNA fingerprinting continues to be indispensable in criminal justice while demanding ongoing interdisciplinary research combining biotechnology, computer science, and law.**

***Index Terms*—DNA fingerprinting, forensic science, crime investigation, STR analysis, PCR, DNA database, biomedical computing, next-generation sequencing, forensic bioinformatics**

I. Introduction

Crime investigation has evolved significantly with advancements in forensic science and biotechnology. Among the various techniques available to forensic investigators, DNA fingerprinting has emerged as one of the most reliable and scientifically defensible methods for determining individual identity in criminal and civil proceedings [1]. Every human being, with the exception of identical (monozygotic) twins, possesses a unique DNA sequence. This biological uniqueness forms the scientific cornerstone of forensic DNA profiling.

Deoxyribonucleic Acid (DNA) is the hereditary material present in nearly all living cells. It carries genetic information from one generation to the next and is primarily located in the nucleus, organised into chromosomes. The DNA double-helix structure consists of two strands joined by nitrogenous base pairs: adenine with thymine, and cytosine with guanine. Although approximately 99.9% of human DNA is identical across individuals, the remaining 0.1% contains sufficient variation at specific genomic loci to distinguish any two people [2].

Since Alec Jeffreys first described genetic fingerprinting at the University of Leicester in 1984, the technique has been applied to criminal identification, paternity testing, disaster victim identification, and immigration disputes [3]. The first forensic application in 1987 solved a double murder case in England and notably both exonerated an innocent suspect and secured a conviction through a mass DNA screening.

A forensic analyst examining a DNA gel profile is shown in Fig. 1. The unique band pattern visible on the gel forms the basis of individual identification in casework.

Modern forensic DNA analysis now integrates automated STR profiling, national DNA databases, computational interpretation tools, and emerging next-generation sequencing platforms. The intersection of computer science and forensic biology has opened new possibilities for faster, more accurate, and privacy-aware DNA identification. This paper presents a survey of these developments with attention to methodology, applications, challenges, and future directions.

II. Literature Survey

DNA fingerprinting has attracted continuous research attention since its inception. Roewer provided a comprehensive review of forensic DNA analysis, tracing the development

Fig. 1. A forensic analyst examining an electrophoresis gel displaying DNA band patterns used for individual identification [1].

of STR typing, Y-chromosome markers, and mitochondrial DNA (mtDNA) analysis as employed in forensic laboratories worldwide [1]. The review highlighted the critical role of national DNA databases in linking crime scene samples to known offenders, a process termed a "cold hit."

Van Oorschot, Ballantyne, and Mitchell examined the significance of trace DNA evidence in criminal investigations, focusing on the ability of modern methods to extract DNA profiles from minimal biological material such as skin cells or faint contact traces [4]. Their work addressed persistent concerns around contamination, secondary transfer, and the probabilistic interpretation of mixed DNA profiles.

Curran and Weir discussed advances in the statistical interpretation of forensic DNA evidence, noting the transition from early gel-based methods to fully automated capillary electrophoresis systems [5]. They emphasised improvements in sensitivity, reproducibility, and the courtroom presentation of likelihood ratios derived from population genetics databases.

Research conducted in the Indian context examined the legal and procedural challenges surrounding DNA evidence admissibility in criminal courts [6]. The study noted that while DNA profiling has become a key part of both criminal and civil proceedings in India, concerns remain regarding database governance, privacy protection, and the standardisation of collection and analysis protocols across forensic laboratories.

The PrivaMatch study introduced privacy-preserving DNA matching frameworks using cryptographic techniques, enabling investigators to compare forensic profiles across institutions without exposing raw genetic data [7]. This work highlights the growing convergence of computer science security research and forensic science.

Collectively, the literature confirms that DNA fingerprinting is scientifically robust but continues to evolve at the intersection of biotechnology, law, and computing. Research gaps remain in automated mixture interpretation, ethical database governance, and deployment of rapid portable DNA analysis systems in field conditions.

III. Methodology

A. DNA Technologies Used in Forensic Analysis

Forensic DNA analysis has employed several successive generations of technology, each improving sensitivity, speed, and discriminatory power.

1) Restriction Fragment Length Polymorphism (RFLP): The classical method uses restriction enzymes to cut DNA at specific sites, producing fragments of varying lengths separated by agarose gel electrophoresis and visualised via Southern blotting with radiolabelled probes [1]. Multi-locus probes detect 15–20 variable fragments per individual. Despite successful application throughout the 1980s and early 1990s, RFLP requires large quantities of high-molecular-weight DNA and is poorly suited to degraded or low-template crime scene samples.

2) Short Tandem Repeat (STR) Analysis: STR analysis, introduced in the early 1990s, became the global standard for forensic DNA profiling. STRs are short repetitive DNA sequences distributed across the genome; the number of repeat units at each locus varies between individuals. PCR amplification allows STR typing from as few as three nucleated cells and from severely degraded material [8]. Up to 30 STR markers can be co-amplified and detected in a single capillary electrophoresis run.

Two core STR standard sets exist: the European Standard Set of 12 markers and the US CODIS set of 13 markers, together forming an international standard of 18 loci [9]. The probability of two unrelated individuals sharing the same profile across 13 STR loci exceeds one in a billion [10].

3) Mitochondrial DNA (mtDNA) Analysis: mtDNA is maternally inherited and present in hundreds to thousands of copies per cell, making it valuable for samples with very low nuclear DNA content such as shed hair shafts, skeletal remains, and ancient specimens [1]. The EMPOP database (www.empop.org) contains approximately 33,000 haplotypes sampled from 63 countries.

4) Y-Chromosome Analysis: The Y chromosome is paternally inherited and non-recombinant, making Y-STR profiling particularly useful in sexual assault cases where an excess of female victim DNA masks the male contributor's autosomal profile [1]. Panels of up to 27 markers are in routine use. The YHRD database (www.yhrd.org) contains approximately 115,000 haplotypes from 850 populations.

B. The Forensic DNA Profiling Workflow

The complete eleven-step process of DNA fingerprinting is illustrated in Fig. 2. The workflow proceeds from biological sample collection through DNA extraction, PCR amplification, electrophoresis, Southern blotting, radioactive probe hybridisation, and X-ray film development to produce a visible DNA fingerprint pattern.

The standard end-to-end analytical process is summarised as follows:

1) **Biological Sample Collection:** Blood stains, semen, saliva, vaginal fluid, hair follicles, bone, and amniotic

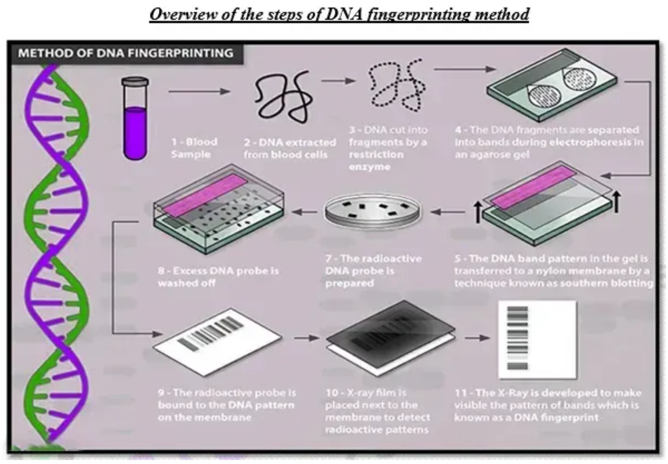

Fig. 2. Overview of the eleven-step DNA fingerprinting method: from blood sample collection to the final X-ray DNA fingerprint pattern [8].

fluid are recovered under strict chain-of-custody protocols.

2) **DNA Extraction:** Biological material is processed to release and purify DNA. Concentration is quantified by spectrophotometry or fluorometry.
3) **PCR Amplification:** As little as 1 nanogram of target DNA is amplified millions of times using thermocyclic PCR with fluorescently labelled primers targeting the STR loci of interest.
4) **Gel Electrophoresis:** Amplified fragments are separated through a polymeric gel-filled glass capillary. A laser excites the fluorescent labels; a CCD camera records excitation peaks at up to 16 loci simultaneously.
5) **Southern Blotting:** The DNA band pattern in the gel is transferred to a nylon membrane for probe hybridisation.
6) **Probe Hybridisation:** A radioactive DNA probe is bound to the DNA pattern on the membrane; excess probe is washed off and the membrane is exposed to X-ray film.
7) **Allele Calling and Profile Generation:** GeneMapper ID software assigns allele calls at each locus, generating a numerical STR profile unique to the individual.
8) **Database Comparison:** The profile is searched against national databases (CODIS in the USA, NDNAD in the UK) to identify matches with previously databased offenders or unsolved case stains.

C. Exhibits Preferred for DNA Examination

Forensic guidelines specify the following preferred biological exhibits:

- Teeth and long bones (femur, humerus) provide robust DNA from skeletal remains.
- Deep muscle tissue or minimally affected skin is preferred in mass disaster scenarios.
- Foetal and maternal tissues must be separated at the point of collection in disputed paternity cases.
- Tissues must **never** be preserved in formalin, which cross-links proteins and severely degrades DNA quality.
- Vaginal swabs in sexual assault cases must be dried and packaged separately.
- Hair roots, if present, must be preserved on slides at the earliest opportunity.

D. Chain of Custody Requirements

Investigating officers submitting exhibits to a forensic DNA laboratory must provide: (1) a brief case history; (2) sealed and marked exhibits with control blood samples; (3) completed Biological Sample Donor Cards in duplicate; (4) wax sample seals; (5) copies of post-mortem or medico-legal reports; (6) a letter of authority addressed to the Director of the Forensic Science Laboratory; (7) chain of custody documentation; and (8) proper packaging and preservation of all biological samples.

IV. Results and Discussion

A. Discriminatory Power of STR Profiling

Modern STR-based DNA profiling achieves exceptional discriminatory power. The probability that two randomly selected, unrelated individuals share identical alleles across all 13 CODIS STR loci is less than one in a billion [10]. Commercial multiplex kits now routinely deliver reproducible profiles from fewer than three nucleated cells. This combination of sensitivity and specificity makes STR profiling uniquely suited to both high-profile crimes and cold cases where biological material has been stored for decades.

B. National DNA Databases and Cold Hits

The establishment of government-controlled criminal DNA databases has fundamentally changed law enforcement capability. The United Kingdom launched the world's first national database in 1995. Table I summarises key global DNA database statistics [1].

TABLE I
Summary of Major National Forensic DNA Databases

Country	DB Name	Profiles (approx.)	% of Pop.
China	National	16 million	∼1.1%
USA	CODIS	10 million	∼3%
UK	NDNAD	6 million	∼10%
Germany	DAD	–	∼0.9%
Netherlands	NDDNA	–	∼0.8%

A "cold hit" — a database match linking a crime scene sample to a previously databased individual — now routinely opens investigative leads in cases that would otherwise remain unsolved. This has been particularly effective in linking serial offenders across multiple unsolved cases.

C. Lineage Markers: Forensic Case Examples

Y-STR profiling demonstrated its value in a 2002 Berlin case in which a woman was attacked in her apartment. Autosomal STR analysis identified two unknown male DNA profiles on items from the scene but could not establish population origin. Court-ordered Y-STR haplotyping and a YHRD database search indicated a Southern European patrilineage, directing investigators to Italy. This led to the identification of both suspects — an uncle and his nephew — demonstrating how biogeographic inference from lineage markers can resolve cases where conventional database searches are inconclusive [1].

The identification of two missing Romanov children provides a landmark example of mtDNA application. Combined mtDNA and Y-STR analysis of skeletal remains near Yekaterinburg provided virtually irrefutable evidence of their identity as the Tsarevich Alexei and one of his sisters, resolving a historical mystery that had persisted since 1918 [1].

D. Familial DNA Searching

Familial searching exploits partial STR profile matches between a crime stain and a databased individual to identify near relatives of the true perpetrator. The first successful familial search was conducted in the UK in 2004, leading to the conviction of Craig Harman for manslaughter via a partial match with his brother's profile [1]. However, this strategy raises significant civil liberties concerns. Following a ruling by the European Court of Human Rights in 2008, the UK destroyed over 1.1 million profiles of innocent individuals. Germany's Federal Constitutional Court ruled against familial searching in 2012.

E. Integration of AI and Next-Generation Sequencing

Next-generation sequencing (NGS) platforms offer the potential to simultaneously analyse autosomal STRs, Y-chromosomal markers, the complete mtDNA genome, ancestry-informative SNPs, and phenotype-predictive variants in a single workflow [1]. Machine learning is being explored for automated mixture deconvolution — the computational separation of DNA profiles from samples containing genetic material from two or more contributors. AI-assisted interpretation systems may reduce analyst subjectivity and improve consistency in evidence reporting.

Commercial rapid DNA instruments can now produce a database-compatible STR profile in approximately two hours using a fully automated "swab in – profile out" process. Twenty-eight US states currently take DNA swabs as part of standard booking procedures following the Supreme Court ruling in *Maryland v. Alonzo Jay King Jr.* (2013).

F. Comparison of Forensic DNA Methods

Table II provides a comparative overview of the main forensic DNA methods discussed in this paper.

TABLE II
COMPARISON OF FORENSIC DNA PROFILING METHODS

Method	Primary Use	Strength	Limitation
RFLP	Historical casework	High specificity	Needs large intact DNA
STR (PCR)	Routine profiling	Sensitive; automated	Mixed profiles complex
Y-STR	Male contributor ID	Works with female-excess DNA	Shared by paternal relatives
mtDNA	Degraded samples, hair	High copy number	Maternal lineage only
NGS	Future full-profile	Multi-target; phenotyping	High error rate currently
AI/ML	Mixture interpretation	Reduces subjectivity	Needs large training data

G. Limitations and Ethical Challenges

Despite its power, forensic DNA analysis faces several persistent challenges. Sample contamination during collection, handling, or laboratory processing can introduce foreign DNA or cause allelic dropout. Degraded or low-template samples may yield incomplete profiles that are difficult to interpret statistically. Laboratory infrastructure for STR analysis remains expensive and inaccessible in many jurisdictions, limiting the global reach of forensic DNA capabilities.

Privacy concerns represent a fundamental societal challenge. Storing genetic profiles of convicted offenders — and in some systems, of arrestees or even all citizens — raises questions about the proportionality of data retention, the risk of misuse, and the potential for genetic discrimination. Legal frameworks for database governance differ substantially between nations, preventing the creation of a unified international forensic database system.

V. CONCLUSION

DNA fingerprinting has fundamentally transformed forensic science and the criminal justice system since its discovery in 1984. Beginning with RFLP-based multi-locus probe methods, the technology has progressed through PCR-based STR analysis to highly automated, database-integrated profiling systems capable of identifying individuals from trace biological material. Lineage markers — Y-STRs and mtDNA — extend the reach of DNA analysis to cases where conventional autosomal STR profiling is insufficient.

The intersection of computer science and forensic biology is driving the next wave of innovation. Artificial intelligence, next-generation sequencing, rapid portable DNA instruments, and privacy-preserving cryptographic frameworks are reshaping the capabilities and ethical boundaries of forensic DNA analysis. Future systems must balance investigative power with robust protections for genetic privacy, data security, and legal due process.

The implementation of advanced techniques such as PCR, STR analysis, mitochondrial DNA analysis, and computerised

DNA databases has enhanced the efficiency and accuracy of forensic investigations. DNA evidence has become a powerful tool in solving complex criminal cases including murder, sexual assault, terrorism, kidnapping, and disaster victim identification. In many cases, DNA analysis has also helped exonerate innocent individuals who were wrongly accused or convicted.

In conclusion, DNA fingerprinting has revolutionised modern forensic science by providing dependable scientific evidence in criminal investigations. The technology continues to play a vital role in maintaining law and order, ensuring justice, protecting innocent individuals, and supporting the advancement of modern criminal investigation systems worldwide.

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